



Estimation Models



Course Instructor

Dr. Sanjeev Jakhar

Associate Professor, Department of Mechanical Engineering



Estimation Model: Time Value for Money

Time Value for Money

a rupee today
is worth more
than a rupee
tomorrow



Do you know why is it so?

XYZ's grandfather decided to give a gift of Rs. One lakh at the end of the fifth year; and gave him a choice of having Rs. 75,000 today. Had you been in XYZ's place what choice would you have made?

- Do you accepted Rs. 1,00,000 after five years or Rs. 75,000 today? What do you say?
- Rs. 75,000 today is much more attractive than Rs. 1,00,000 after five years because the **present is more certain than the future**. You could **invest** Rs. 75,000 **in the market** and earn a return on this amount. Rs. 1,00,000 at the end of five years would have less purchasing power due to inflation.



Estimation Model: **Payback Period**

Definition

The payback period refers to the **length of time** it takes to **recover the cost of an initial investment**.

Scope

- It helps investors and managers to make a **quick judgments or economic-decision making** on their **investments** through use the payback period.
- An investment project with a short payback period promises the **quick cash inflow**. It is therefore, a useful capital budgeting method for cash poor firms.

Methods for Estimation of Payback Period

- Even cash flow
- Un-even cash flow

Pros:

- It is very simple and cost effective technique.
- It measures direct relationship between annual cash flows from proposal and the net investment required.

Cons:

- The payback period does not account for what happens after payback, **ignoring the overall profitability of an investment**.
- One of the downsides of the payback period is that **it disregards the time value of money**.



Payback Period: Even cash flow

- When net annual cash inflow is even (i.e., **same cash flow every period**).
- The payback period of the project can be computed by applying the simple formula given below:

$$\text{➤ Payback period} = \frac{\text{Initial Investment}}{\text{Annual Cash Flow}}$$

- Company X is planning to undertake a project requiring initial cost of \$10500 the project is expected to generate \$2500 per year for 7 years. Calculate the payback period of project.

$$\begin{aligned}\text{Payback period} &= \frac{\$10500}{\$2500} \\ &= 4.2 \text{ Years}\end{aligned}$$



Even cash flow: Numerical

➤ Due to increased demand, the management of XYZ Beverage Company is considering to purchase a new equipment to increase the production and revenues. The useful life of the equipment is **10 years** and the company's maximum **desired payback period is 4 years**. The inflow and outflow of cash associated with the new equipment is given below:

- Initial cost of equipment: \$37,500
- Annual cash inflows: Sales: \$75,000
- Annual cash Outflows:
 - Cost of ingredients: \$45,000
 - Salaries expenses: \$13,500
 - Maintenance expenses: \$1,500

Should XYZ Beverage Company purchase the new equipment? Make a decision-making using payback method.



Even cash flow: Solution

Step 1:

- In order to compute the payback period of the equipment, we need to work out the net annual cash inflow by deducting the total of cash outflow from the total of cash inflow associated with the equipment.

$$\begin{aligned}\text{Computation of net annual cash inflow} &= \text{cash inflow} - \text{cash outflow} \\ &= \$75,000 - (\$45,000 + \$13,500 + \$1,500) \\ &= \$15,000\end{aligned}$$

Step 2:

$$\text{Payback period} = \frac{\text{Initial Investment}}{\text{Annual Cash Flow}}$$

$$\text{Payback period} = \frac{\$37500}{\$15000} = 2.5 \text{ Years}$$

So, the equipment should be purchased because the payback period of the equipment is 2.5 years which is shorter than the maximum desired payback period of 4 years.



Payback Period: Un-even cash flow

- When projects generate inconsistent or uneven cash inflow (**different cash inflow in different periods**), the simple formula given above cannot be used to compute payback period.
- In such situations, we need to compute the cumulative cash inflow and then apply the following formula:

➤ *Payback period =*

$$\text{Years before full recovery} + \frac{\text{Unrecovered cost at start of the year}}{\text{Cash flow during the year}}$$



Payback Period: Un-even cash flow

➤ An investment of \$200,000 is expected to generate the following cash inflows in six years:

- Year 1: \$70,000
- Year 2: \$60,000
- Year 3: \$55,000
- Year 4: \$40,000
- Year 5: \$30,000
- Year 6: \$25,000

Compute payback period of the investment. Should the investment be made if management wants to recover the initial investment within 3 years or less?



Payback Period: Un-even cash flow

Year	Cash inflow	Cumulative cash inflow
1	\$ 70,000	= \$ 70,000
2	\$ 60,000	= \$ 130,000
3	\$ 55,000	= \$ 185,000
4	\$ 40,000	= \$ 225,000
5	\$ 30,000	= \$ 255,000
6	\$ 25,000	= \$ 280,000

*Unrecovered investment at start of 4th year:
= Initial cost – Cumulative cash inflow at the end of 3rd year
= \$200,000 – \$185,000
= \$15,000

➤ *Payback period = Years before full recovery + $\frac{\text{Unrecovered cost at start of the year}}{\text{Cash flow during the year}}$*

➤ *Payback period = $3 + \frac{\$ 15,000}{\$ 40,000} = 3 + 0.375 = 3.375$ Years*



Payback Period: Un-even cash flow

Example:-Initial investment: Rs. 50000. It generates a cash flow Rs 15000, Rs 20000, Rs 14000, Rs 16000 during the next 4 years. Find out the payback period.

Year		Cash flow
0	-50000	
1	-35000	15000
2	-15000	20000
3	-1000	14000
4	15000	16000



$$\begin{aligned}\text{Payback} &= 3 + \frac{1000}{16000} \\ &= 3.0625\end{aligned}$$



Rate of Return (ROR)

Definition

- It represents the annual return on the investment, **it is the reciprocal of simple payback.**

$$ROR = \frac{\text{Saving/Year}}{\text{Initial Investment}}$$

- Calculate the ROR, if a company X is planning to undertake a project requiring initial investment of \$10500 the project is expected to generate \$2500 per year for 7 years. Calculate the ROR of project.

$$ROR = \frac{\$ 2500}{\$ 10500}$$

$$= .2380 = 23.80 \%$$



Future Value (F)

Definition

- It is the **value of a current asset** at a **future date** based on an assumed **rate** of growth.
- The future value is important to investors and financial planners, as they use it to estimate **how much an investment made today will be worth in the future.**

Future value calculations

- The “future value of a lump sum” is the value of a single deposit, **like a bank fixed deposit over time.**
- The “future value of an annuity” is the value of a series of payments, **like constant payment of insurance premium at regular intervals, over time.**

- **Simple Annual Interest:**
- If an investment earns simple interest, then the FV formula is:

$$F = P \times (1 + i \times n)$$

- **Compounded Annual Interest :**
- if rate is applied to each period's cumulative account balance, then FV formula is:

$$F = P \times (1 + i)^n$$

Where,

P = Present amount; n = Number of years

F = Future value; i = Interest rate



Future Value and Present Value Calculation (F)

- Consider, present value ($P = 100$); **rate of interest** ($i = 5\%$); over (n) compounding years. The future value (F) can be determined as:

$$F_0 = P = 100$$

$$F_1 = P + P(i) = P(1 + i) = 100 + 100 \times 0.05$$

$$F_2 = P(1 + i) + P(1 + i)(i) = P(1 + i)[1 + i]$$

$$F_2 = P(1 + i)^2 = 100(1 + 0.05)^2$$

$$F_n = P(1 + i)^n$$

- This can be rearranged to show the present value of a future amount (**Present value of a future amount**):

$$P = \frac{F}{(1 + i)^n}$$

- The factor $(1 + i)^{-n}$ or $\frac{1}{(1 + i)^n}$ is sometimes called the **present worth factor (PWF)**.

➤ Someone promises to pay you \$1000 in 5 years. If the interest rate is 10% per year, what amount would you take today that is equivalent to \$1000 in 5 years.

$$P = \frac{F}{(1 + i)^n}$$

$$P = \frac{1000}{(1 + 0.1)^5}$$

$$P = \$621$$



Net Present Value of Series of Cash in-flows (NPV)

INTRODUCTION

- Net present value (NPV) is the difference between the present value of cash inflows and the present value of cash outflows over a period of time. It may be positive, zero or negative.
- NPV is used in capital budgeting and investment planning to analyze the profitability of a projected investment or project.
- Also known as sophisticated technique for capital budgeting exercise.
- It accounts for time value of money by using discounted cash flows in the calculation.



Net Present Value of Series of Cash in-flows (NPV)

- Net present value (NPV) is the difference between the present value of **cash inflows** and the present value of **cash outflows** over a period of time.
- $F_n = A = F_n$ is *Cash inflows in respective year n .*

When cash inflows is even or uneven

$$NPV = \sum_{n=1}^{n=t} \frac{F_n}{(1+i)^n} - P \text{ (Initial investment (cash outflow at time 0))}$$



Net Present Value of Series of Cash in-flows (NPV)

When cash inflows is even

$$P_0 = F_0$$

$$P_1 = \frac{F_1}{(1+i)^1} = \frac{A}{(1+i)^1} \quad \text{If you receive ₹A after 1 year}$$

After 2 years

$$P_2 = \frac{A}{(1+i)^2}$$

After 3 years

$$P_3 = \frac{A}{(1+i)^3}$$

$$P_{n=t} = \frac{A}{(1+i)^t}$$

If every year you receive **same amount A (Annuity case)**, then:

$$P_n = A \times \left[\frac{1}{(1+i)^n} + \frac{1}{(1+i)^{n-1}} + \frac{1}{(1+i)^{n-2}} + \dots + \frac{1}{(1+i)^2} + \frac{1}{(1+i)^1} \right]$$



Net Present Value of Series of Cash in-flows (NPV)

$$S_n = a + ar + ar^2 + ar^3 + \dots + ar^{n-1}$$

$$S_n = a[(1-r^n)/(r-1)]$$

$$\text{First term} = a = \frac{A}{(1+i)}$$

$$\text{Common factor} = r = \frac{1}{(1+i)}$$

For geometric progression series

$$P_n = A \times \left[\frac{1 - (1+i)^{-n}}{i} \right]$$

Its calculate Present Value (PV) of cash inflows only

NPV = PV of inflows – Initial Investment

$$\text{NPV} = A \times \left[\frac{1 - (1+i)^{-n}}{i} \right] - P$$



Net Present Value Calculation (NPV)

Example:- Someone promises to pay you \$1000 in 5 years. If the interest rate is 10% per year, what amount would you take today that is equivalent to \$1000 in 5 years?

$$P = F / (1+i)^n = 1000 / (1+0.1)^5$$

NPV > 0 Accept the project

NPV < 0 Reject the project

NPV = 0 Accept or Reject the project

The calculation of NPV will result in **three possible outcomes**:

- **Positive NPV:** In this situation, the present value of cash inflows **is greater than** the present value of cash outflows. **This is an ideal situation for investment**
- **Negative NPV:** In this situation, the present value of cash inflows **is less than** the present value of cash outflows. This is not an ideal situation, and any project with this NPV should not be accepted.
- **Zero NPV:** In this situation, the present value of cash inflows equals the present value of cash outflows. You may or may not accept the project.



Net Present Value Calculation (NPV)

EXAMPLE

Let's assume Company X wants to buy Company A. It takes a careful look at Company A's projections for the next 10 years. It discounts those projected cash inflows back to the present using its weighted average cost of capital (WACC) and then subtracts the cost of purchasing Company A.



Net Present Value Calculation (NPV)

Cost to purchase Company A : **Rs.10,00,000**

Present value (PV) of cash flows from acquiring Company A:

Year 1: Rs.200,000

Year 2: Rs.150,000

Year 3: Rs.100,000

Year 4: Rs.75,000

Year 5: Rs.70,000

Year 6: Rs.55,000

Year 7: Rs.50,000

Year 8: Rs.45,000

Year 9: Rs.30,000

Year 10: Rs.10,000

Total: Rs.7,85,000

Net Present Value (NPV) = Rs.785,000 - Rs.10,00,000 = **-Rs.215,000**

At this point, management for Company X would use the net present value rule to decide whether or not to pursue the acquisition of Company A. Because the **NPV is negative**, they should say, "No."



Net Present Value Calculation (NPV)

Problem: Suppose a project requires an **initial investment of \$2000** and it is expected to generate a **cash flow of \$100 for 3 years** plus **\$2500 in the third year**. The **target rate of return of the project is 10% per annum**. Calculate the **net present value** of the project.

Solution:

Cash outlay = Initial investment = $F_0 = P = \$2000$

Discounted rate of return = $i = 10\%$

$F_1 = \$100; F_2 = \$100; F_3 = \$100;$

$$NPV = \sum_{n=1}^{n=t} \frac{F_n}{(1+i)^n} - \text{cash outlay } (F_0 = P)$$

$$NPV = \frac{\$100}{(1+0.10)^1} + \frac{\$100}{(1+0.10)^2} + \frac{\$100}{(1+0.10)^3}$$

$$+ \frac{\$2500}{(1+0.10)^3} - \$2000$$

$$NPV = \$90.91 + \$82.64 + \$75.13$$

$$+ \$1878.29 - \$2000$$

At 10% discount rate, total present value of all future cash flows

$$NPV = \$126.97 \quad NPV > 0 \quad (\text{Accept the project})$$

Therefore, NPV of the project at 10% is \$126.97.



Net Present Value Calculation (NPV)

Problem: Assume that ABC Inc is considering two projects namely Project X and Project Y and wants to calculate the NPV for each project. Both project X and project Y is four-year project and cash flows of both the projects for four years are given below:

Year	Project A Cash Flows	Project B Cash Flows
1.	\$ 5000	\$ 1000
2.	\$ 4000	\$ 3000
3.	\$ 3000	\$ 4000
4.	\$ 1000	\$ 6750

The firm's cost of capital is 10% for each project and the initial investment amount is \$10,000. Calculate the NPV of each project and determine in which project the firm should invest.



Net Present Value Calculation (NPV)

Solution:

$$NPV = \sum_{n=1}^{n=t} \frac{F_n}{(1+i)^n} - \text{cash outlay } (F_0 = P)$$

Year	Project A cash in-flows	PV	Project B cash in-flows	PV
1	\$5000	$\frac{\$5000}{(1+0.10)^1} = \4545.45	\$1000	$\frac{\$1000}{(1+0.10)^1} = \909.09
2	\$4000	$\frac{\$4000}{(1+0.10)^2} = \3305.78	\$3000	$\frac{\$3000}{(1+0.10)^2} = \2479.33
3	\$3000	$\frac{\$3000}{(1+0.10)^3} = \2253.94	\$4000	$\frac{\$4000}{(1+0.10)^3} = \3053.43
4	\$1000	$\frac{\$1000}{(1+0.10)^4} = \683.01	\$6750	$\frac{\$6750}{(1+0.10)^4} = \4610.34
		$NPV = \$10780.18 - \10000 $NPV = \$780.18$		$NPV = \$11052.19 - \10000 $NPV = \$1052.19$

$NPV_B > NPV_A$ (Accept the project B)

Therefore, NPV of the project B is higher compared to project A, therefore Project B is better for investment.



Net Present Value Calculation (NPV)

Qn. A standard-efficiency furnace costs \$100 and consumes \$40 per year in fuel over a 10 year lifetime. A high efficiency furnace costs \$200 and consumes \$20 per year in fuel over a 10 year lifetime. If the interest rate is 10% per year, which is better investment?

$$P_s = \$100(\text{Initial cost}) + \$40(\text{annual cost}) \left[\frac{1 - (1+0.1)^{-10}}{0.1} \right] = \$346 \dots\dots\dots C_0 + A \left(\frac{1 - (1+i)^{-n}}{i} \right)$$

$$P_h = \$200 + \$20 \left[\frac{1 - (1+0.1)^{-10}}{0.1} \right] = \$323$$

Saving = \$346 - \$323 = \$ 23 High efficiency furnace is better investment.

Or directly

$$P_{h, \text{savings}} = -\$100 + \$20 \left[\frac{1 - (1+0.1)^{-10}}{0.1} \right] = \dots\dots\dots$$



Net Present Value Calculation (NPV)

- Effect of discount rate
 - The discount rate is the expected rate of return from alternate investment
 - High discount rates reflects the belief that a large profit can be made from an alternative investment
- In the previous Question, if the discount rate is 30% instead of 10%

$$P_{h, savings} = -\$100 + \$20 \left[\frac{1 - (1+0.3)^{-10}}{0.3} \right] = \dots\dots\dots$$

- It indicates that at high discount rate, the standard furnace with low initial cost is the better investment.



Net Present Value Calculation (NPV)

Qn. Using payback period and NPV methods, evaluate the following projects and select the project that should be undertaken. Assume a discount rate of 10%.

Project A (\$)	Project B (\$)	Year
-1200	-1200	0
400	400	1
400	400	2
400	350	3
200	800	4
200	800	5

A – Payback period = 3 years

B – Payback period = $3 + \frac{50}{800} = 3.0625 \text{ Years}$



Net Present Value Calculation (NPV)

Net present value, NPV for project A

$$\begin{aligned} &= -1200 + \frac{400}{(1+0.1)} + \frac{400}{(1+0.1)^2} + \frac{400}{(1+0.1)^3} + \frac{200}{(1+0.1)^4} + \frac{200}{(1+0.1)^5} \\ &= 55 \end{aligned}$$

NPV for project B

$$\begin{aligned} &= -1200 + \frac{400}{(1+0.1)} + \frac{400}{(1+0.1)^2} + \frac{350}{(1+0.1)^3} + \frac{800}{(1+0.1)^4} + \frac{800}{(1+0.1)^5} \\ &= 800.32 \end{aligned}$$



Net Present Value Calculation (NPV)

- **Practice problem:**

Consider investing in a real-estate project. Purchasing the land Rs. 200000. The project will generate a cash flow of Rs. 30000/year for next 3 years. Cost of capital is 12%. Calculate the NVP and determine it is financially viable or not.

$$P_n = A \left[\frac{1 - (i + 1)^{-n}}{i} \right]$$

$$P_n = 30000 \left[\frac{1 - (0.12 + 1)^{-3}}{0.12} \right] = 72054 \text{ Rs.}$$

$$\text{NVP} = -200000 + 72054 = -127946 \text{ Rs.}$$

$\text{NVP} < 0$; the project is not viable financially.



Internal Rate of Return (IRR)

- IRR, or internal rate of return, is a metric used in financial analysis **to estimate the profitability of potential investments.**
- IRR is a **discount rate** that makes the **net present value (NPV)** of all cash flows equal to zero in a **discounted cash flow analysis.** Or IRR is the rate of return at which the present value of cash inflows equals the present value of cash outflows.
- **IRR calculations rely on the same formula as NPV does.** Keep in mind that IRR is not the actual dollar value of the project. It is the annual return that makes the NPV equal to zero.
- The ultimate goal of **IRR is to identify the rate of discount**, which makes the present value of the sum of annual nominal cash inflows equal to the initial net cash outlay for the investment.
- IRR is ideal for analysing capital budgeting projects to understand and compare potential rates of annual return over time.



How is NPV different from IRR?

- **NPV** is the difference between the present value of cash inflows and the present value of cash outflows.
- Internal Rate of Return (IRR) on the other hand, is the discount rate that equates the present value of cash inflows with the present values of cash outflows.
- IRR is calculated by assuming that NPV is equal to zero.
- It shows the breakeven point or the point where there's neither profit nor loss.
- Conversely, NPV shows the excess inflows over the outflows.



The Formula for IRR

The discount rate which sets the **NPV of all cash flow equal to zero**. It helps to determine the yield on an investment (**For uneven cash flow**).

$$NPV = -Initial\ investment + \frac{cash\ flow}{(1 + IRR)^1} + \dots + \frac{CF_n}{(1 + IRR)^n}$$

Where

F_n = Net cash in-flow during the period

P = Total net cash outflow during the period or initial investment

IRR = The internal rate of return

n = The number of time periods

$$0 = NPV = \sum_{n=1}^{n=t} \frac{F_n}{(1 + IRR)^n} - P ; \quad (F_0 = P)$$

This equation cannot be solved directly by algebra (for large n).

We use:

- Trial and Error
- **Interpolation**
- Financial calculator
- Excel (IRR function)



The Formula for IRR

IRR is calculated by the **interpolation formula**

$$i_* = i_1 + \frac{i_2 - i_1}{y_2 - y_1}(y_* - y_1)$$

Suppose a project gives:

•NPV at **10%** = **+200**

•NPV at **15%** = **-100**

We want to find IRR.

$$i_1 = 10\%, \quad y_1 = 200$$

$$i_2 = 15\%, \quad y_2 = -100$$

$$y_* = 0$$

$$i_* = 10 + \frac{15 - 10}{-100 - 200}(0 - 200)$$

$$i_* = 13.33\%$$

Where:

- i_* = required rate (IRR)
- i_1 = rate giving positive NPV
- i_2 = rate giving negative NPV
- y_1 = NPV at i_1
- y_2 = NPV at i_2
- $y_* = 0$ (because IRR makes NPV zero)



The Formula for IRR

Example: The cash flows associated with a machine are as follows: Initial cost (F_0)=-\$50,000 and additional cost is (F_j) = \$15,000 for 5 years ($j = 1, \dots, 5$). Determine the economic acceptability of this machine at interest rates of (a) 10%, (b) 15%, and (c) 20% per year. Also find the IRR of project.

$$0 = NPV = \sum_{n=1}^{n=t} \frac{F_n}{(1 + IRR)^n} - P ; \quad (F_0 = P)$$

Initial investment: $F_0 = -50,000$

Annual cash inflow: $F_j = 15,000$ ($j = 1$ to 5)

Project life: $n = 5$ years

We will use **NPV method** to check acceptability.

(a) $NPV = -\$50,000 + \$15,000(3.7908) = \$6,861.80$ (Economically acceptable investment)

(b) $NPV = -\$50,000 + \$15,000(3.3522) = \$282.33$ (Barely acceptable)

(c) $NPV = -\$50,000 + \$15,000(2.9906) = -\$5,140.82$ (Not economically justifiable)



The Formula for IRR

By linear interpolation between the results of (b) and (c):

i (%)	NPV (\$)
15	282.33
i^*	0
20	-5,140.82

$$i^* = 15\% + \left[\frac{0 - 282.33}{-5140.82 - 282.33} \right] * [20\% - 15\%]$$
$$i^* = 15.26\%$$

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IRR is the rate at which: $NPV = 0$

So to use interpolation, we **must choose two rates where:**

- One NPV is **positive**
- One NPV is **negative**

Because IRR lies **between a positive and a negative NPV.**



Internal Rate of Return (IRR)

Example:- A company is deciding whether to purchase new equipment that costs 500000. Management estimates the life of the new asset to be four years and expects it to generate an additional 160000 of annual profit. In the fifth year, the company plans to sell the equipment for its salvage value of 50000. Calculate the IRR.

Assuming the ₹50,000 salvage value is received at the end of the 4th year, the cash flows are:

Year	Cash Flow (₹)
0	-5,00,000
1	+1,60,000
2	+1,60,000
3	+1,60,000
4	+2,10,000

The IRR is obtained from:

$$500000 = \frac{160000}{(1+r)} + \frac{160000}{(1+r)^2} + \frac{160000}{(1+r)^3} + \frac{210000}{(1+r)^4}$$

Solving this equation:

$$IRR \approx 13.61\%$$

Or

At 13%:

$$NPV_{13} = -500000 + \frac{160000}{1.13} + \frac{160000}{1.13^2} + \frac{160000}{1.13^3} + \frac{210000}{1.13^4}$$
$$NPV_{13} \approx +\text{₹}9,635$$

At 14%:

$$NPV_{14} = -500000 + \frac{160000}{1.14} + \frac{160000}{1.14^2} + \frac{160000}{1.14^3} + \frac{210000}{1.14^4}$$
$$NPV_{14} \approx -\text{₹}6,774$$

So, IRR lies between 13% and 14%.

$$IRR \approx 13.6\%$$